

Differential Calculus

*Home work sheet # 10

Curl, Divergence and Gradient

17–22 Find $\text{div } \mathbf{F}$ and $\text{curl } \mathbf{F}$. ■

17. $\mathbf{F}(x, y, z) = x^2\mathbf{i} - 2\mathbf{j} + yz\mathbf{k}$

18. $\mathbf{F}(x, y, z) = xz^3\mathbf{i} + 2y^4x^2\mathbf{j} + 5z^2y\mathbf{k}$

19. $\mathbf{F}(x, y, z) = 7y^3z^2\mathbf{i} - 8x^2z^5\mathbf{j} - 3xy^4\mathbf{k}$

20. $\mathbf{F}(x, y, z) = e^{xy}\mathbf{i} - \cos y\mathbf{j} + \sin^2 z\mathbf{k}$

21. $\mathbf{F}(x, y, z) = \frac{1}{\sqrt{x^2 + y^2 + z^2}}(x\mathbf{i} + y\mathbf{j} + z\mathbf{k})$

22. $\mathbf{F}(x, y, z) = \ln x\mathbf{i} + e^{xyz}\mathbf{j} + \tan^{-1}(z/x)\mathbf{k}$

23–24 Find $\nabla \cdot (\mathbf{F} \times \mathbf{G})$. ■

23. $\mathbf{F}(x, y, z) = 2x\mathbf{i} + \mathbf{j} + 4y\mathbf{k}$

$\mathbf{G}(x, y, z) = x\mathbf{i} + y\mathbf{j} - z\mathbf{k}$

33–40 Find ∇z or ∇w . ■

33. $z = \sin(7y^2 - 7xy)$

34. $z = 7 \sin(6x/y)$

35. $z = \frac{6x + 7y}{6x - 7y}$

36. $z = \frac{6xe^{3y}}{x + 8y}$

37. $w = -x^9 - y^3 + z^{12}$

38. $w = xe^{8y} \sin 6z$

39. $w = \ln \sqrt{x^2 + y^2 + z^2}$

40. $w = e^{-5x} \sec x^2 yz$

41–46 Find the gradient of f at the indicated point. ■

41. $f(x, y) = 5x^2 + y^4$; $(4, 2)$

42. $f(x, y) = 5 \sin x^2 + \cos 3y$; $(\sqrt{\pi}/2, 0)$

43. $f(x, y) = (x^2 + xy)^3$; $(-1, -1)$

44. $f(x, y) = (x^2 + y^2)^{-1/2}$; $(3, 4)$

*These problems are for the students only as home work. Search the reference books for more examples.